

Creative MCQ - Way to Success (10th Maths)

Chapters 1 & 8 FULL SOLUTION

Chapter 1: Relations and Functions

Question

1. Let $f(x) = x^2 - x$ then $f(x-1) - f(x+1)$ is

- (A) $4x$
- (B) $2 - 2x$
- (C) $2 - 4x$
- (D) $4x - 2$

Solution

Given $f(x) = x^2 - x$

$$f(x-1) = (x-1)^2 - (x-1) = x^2 + 1 - 2x - x + 1 = x^2 - 3x + 2$$

$$f(x+1) = (x+1)^2 - (x+1) = x^2 + 1 + 2x - x - 1 = x^2 + x$$

$$f(x-1) - f(x+1) = (x^2 - 3x + 2) - (x^2 + x) = 2 - 4x$$

Answer: (C)

Question

2. Given $f(x) \equiv (-1)^x$ is a function from $\mathbb{N}$ to $\mathbb{Z}$. Then the range of f is

- (A) $\{1\}$
- (B) $\mathbb{N}$
- (C) $\{1, -1\}$
- (D) $\mathbb{Z}$

Solution

$$\text{For } x \in \mathbb{N}, f(x) = (-1)^x = \begin{cases} -1 & \text{if } x \text{ is odd} \\ 1 & \text{if } x \text{ is even} \end{cases}$$

$$\therefore f(x) = (-1)^x = \{1, -1\}$$

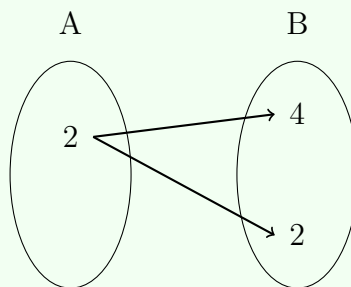
Answer: (C)

Question

3. The given diagram represents

- (A) an onto function
- (B) a constant function
- (C) an one-one function
- (D) not a function

Solution



2 has two images 4, 2. It is not a function.

Answer: (D)

Question

4. $f = \{(2, a), (3, b), (4, b), (5, c)\}$ is a

- (A) identity function
- (B) one-to-one function
- (C) many-one function
- (D) constant function

Solution

Elements 3 and 4 have the same image b . Therefore, it is a many-one function.

Answer: (C)

Question

5. If $f : A \rightarrow B$ is a constant function, then the range of f will have _____ elements.

- (A) 0
- (B) 1
- (C) 2
- (D) 10

Solution

A function $f : A \rightarrow B$ is a constant function then if the range of f contains only one element. i.e., $f(x) = c$, for all $x \in A$ for some fixed $c \in B$.

Answer: (B)

Question

6. The domain of the function $f(x) = \frac{1}{x(x+1)}$ is

- (A) $\{0, -1\}$
- (B) $\mathbb{R} - \{0, -1\}$
- (C) $\mathbb{R} - \{0\}$
- (D) $\mathbb{R} - \{-1\}$

Solution

$$f(x) = \frac{1}{x(x+1)}$$

$$x(x+1) \neq 0 \implies x \neq 0, x \neq -1$$

$$\text{Domain} = \mathbb{R} - \{0, -1\}$$

Answer: (B)

Question

7. If $\{(5, 1), (6, b)\}$ represents a constant function, then the value of b is

- (A) 5
- (B) 6
- (C) 7
- (D) 1

Solution

$\{(5, 1), (6, b)\}$ represents a constant function. So the image must be the same.

$$b = 1$$

Answer: (D)

Question

8. For any two sets P and Q , $P \cap Q$ is

- (A) $\{x : x \in P \text{ or } x \in Q\}$
- (B) $\{x : x \in P \text{ and } x \notin Q\}$
- (C) $\{x : x \in P \text{ and } x \in Q\}$
- (D) $\{x : x \notin P \text{ and } x \in Q\}$

Solution

By the definition of intersection of sets, $P \cap Q = \{x : x \in P \text{ and } x \in Q\}$.

Answer: (C)

Question

9. Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$ then $B \times A$

- (A) $\{(1, a), (1, b), (2, a), (2, b)\}$
- (B) $\{(1, a), (2, a), (2, b)\}$
- (C) $\{(a, b), (a, 2), (b, 1), (b, 2)\}$
- (D) $\{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$

Solution

$$B \times A = \{a, b\} \times \{1, 2, 3\} = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

Answer: (D)

Question

10. $n(A) = 3$ and $n(B) = 4$ then $n(B \times A)$ is

- (A) 5
- (B) 7
- (C) 6
- (D) 12

Solution

$$n(A) = 3 \text{ and } n(B) = 4$$

$$n(B \times A) = n(B) \times n(A) = 4 \times 3 = 12$$

Answer: (D)

Question

11. Real numbers $\mathbb{R}$ is defined as follows

- (A) $\mathbb{N} \cup \mathbb{W}$
- (B) $\mathbb{Q} \cup \mathbb{Q}'$
- (C) $\mathbb{W} \cup \mathbb{Z}$
- (D) $\mathbb{Z} \cup \mathbb{Q}$

Solution

Real numbers are the union of rational ($\mathbb{Q}$) and irrational ($\mathbb{Q}'$) numbers.

Answer: (B)

Question

12. $A = \{1, 2\}$ and $B = \{1, 2\}$ then geometrically the set of points of $A \times B$ is

- (A) a rectangle
- (B) a square
- (C) a circle
- (D) a triangle

Solution

$$A = \{1, 2\}, B = \{1, 2\}$$

$$A \times B = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

Plotting these coordinates on a graph forms a square of side length 1.

Answer: (B)

Question

13. If $B \times A = \{(1, 2), (1, 3), (3, 2), (3, 3), (5, 2), (5, 3)\}$ then find A

- (A) $\{2, 3\}$
- (B) $\{1, 2, 3, 5\}$
- (C) $\{1, 3, 5\}$
- (D) $\{\}$

Solution

A is the set of all second coordinates of elements of $B \times A$.

$$A = \{2, 3\}$$

Answer: (A)

Question

14. $A \times B = \{(3, 2), (3, 4), (5, 2), (5, 4)\}$ then find $A \cup B$

- (A) $\{3, 5\}$
- (B) $\{2, 3, 4, 5\}$
- (C) $\{2, 4\}$
- (D) $\{2, 3, 5\}$

Solution

$$A = \{3, 5\}, B = \{2, 4\}$$

$$A \cup B = \{2, 3, 4, 5\}$$

Answer: (B)

Question

15. $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$ which of the following are relations from B to A ?

- (A) $\{(c, a), (c, b), (c, 1)\}$
- (B) $\{(a, 4), (b, 3), (c, 2)\}$
- (C) $\{(1, a), (b, 4), (c, 3)\}$
- (D) $\{(1, b), (1, c), (3, a), (4, b)\}$

Solution

A relation from B to A must have elements in the form (b, a) where $b \in B$ and $a \in A$. Option B is the only set that strictly maps letters from B to numbers in A .

Answer: (B)

Question

16. $f(x) = \frac{1}{1+x}$ f is defined for all real numbers except at

- (A) $x = 1$
- (B) $x = 0$
- (C) $x = -1$
- (D) $x = 2$

Solution

$$1 + x \neq 0 \implies x \neq -1$$

f is defined for all real numbers except at $x = -1$.

Answer: (C)

Question

17. $f(x) = \frac{1}{x^2 - 5x + 6}$ then domain of f is

- (A) $\mathbb{R} - \{2, 3\}$
- (B) $\mathbb{R}$
- (C) $\mathbb{N}$
- (D) $\mathbb{Z}$

Solution

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

At $x = 2, 3$ then $f(2)$ and $f(3)$ are not defined.

Hence f is defined for all real numbers except at $x = 2$ and 3 .

So domain of $f = \mathbb{R} - \{2, 3\}$

Answer: (A)

Question

18. $f(x) = 2x - x^3$, find $f(-1)$

- (A) 1
- (B) -1
- (C) 3
- (D) -3

Solution

$$f(-1) = 2(-1) - (-1)^3 = -2 - (-1) = -2 + 1 = -1$$

Answer: (B)

Question

19. If a function $f : A \rightarrow B$ is both one-one and onto then f is called

- (A) Injection
- (B) Surjection
- (C) Bijection
- (D) into function

Solution

By definition, a function that is both injective (one-one) and surjective (onto) is bijective.

Answer: (C)

Question

20. If $f(x) = x^m$ and $g(x) = x^n$ then $f \circ g =$

- (A) x^{m+n}
- (B) x^{m-n}
- (C) x^{mn}
- (D) 1

Solution

$$f \circ g = f(g(x)) = f(x^n) = (x^n)^m = x^{mn}$$

Answer: (C)

Chapter 2: Numbers and Sequences

Question

1. The next term of the sequence $\frac{1}{2}, \frac{1}{6}, \frac{1}{10}, \frac{1}{14}, \dots$ is

- (A) $\frac{1}{15}$
- (B) $\frac{1}{16}$
- (C) $\frac{1}{18}$
- (D) $\frac{1}{20}$

Solution

Given $\frac{1}{2}, \frac{1}{6}, \frac{1}{10}, \frac{1}{14}, \dots$

The denominators are forming an A.P: 2, 6, 10, 14, ... with common difference $d = 4$.

Next denominator = $14 + 4 = 18$.

Next term is $\frac{1}{18}$.

Answer: (C)

Question

2. If t_n is the n^{th} term of an A.P., then $t_{8n} - t_n$ is

- (A) $(8n - 1)d$
- (B) $(8n - 2)d$
- (C) $(7n - 2)d$
- (D) $(7nd)$

Solution

$$\begin{aligned} t_{8n} - t_n &= [a + (8n - 1)d] - [a + (n - 1)d] \\ &= a + (8n - 1)d - a - (n - 1)d \\ &= (8n - 1)d - (n - 1)d \\ &= (8n - 1 - n + 1)d = (7n)d \end{aligned}$$

Answer: (D)

Question

3. The sequence $-3, -3, -3, \dots$ is

- (A) An A.P only
- (B) a G.P only
- (C) Neither A.P nor G.P
- (D) both A.P and G.P

Solution

A constant sequence has common difference $d = 0$ and common ratio $r = 1$. Thus, it is both an A.P and a G.P.

Answer: (D)

Question

4. If $2 + 4 + 6 + \cdots + 2k = 90$, then the value of k is

- (A) 8
- (B) 9
- (C) 10
- (D) 11

Solution

$$2(1 + 2 + 3 + \cdots + k) = 90$$

$$2 \left(\frac{k(k+1)}{2} \right) = 90$$

$$k(k+1) = 90$$

$$k^2 + k - 90 = 0$$

$$(k-9)(k+10) = 0 \implies k = 9 \text{ or } k = -10$$

Since k must be positive, $k = 9$.

Answer: (B)

Question

5. If a and b are two positive integers where $a > 0$ and b is a factor of a , then HCF of a and b is

- (A) b
- (B) a
- (C) $3ab$
- (D) $\frac{a}{b}$

Solution

Since b is a factor of a , b divides a completely. The highest common factor is b itself.

Answer: (A)

Question

6. If a, b, c are in A.P then $\frac{a-b}{b-c}$ is equal to

- (A) $\frac{a}{b}$
- (B) $\frac{b}{c}$
- (C) $\frac{a}{c}$
- (D) 1

Solution

Since a, b, c are in A.P, the common differences are equal.

$$b - a = c - b \implies -(a - b) = -(b - c) \implies a - b = b - c$$

$$\therefore \frac{a-b}{b-c} = 1$$

Answer: (D)

Question

7. Sum of the series $2 + 2 + 2 + \dots$ upto n -terms is

- (A) 2^n
- (B) $2n$
- (C) n^2
- (D) $n + 2$

Solution

Given, $2 + 2 + 2 + \dots + n$ terms

$$= 2(1 + 1 + 1 + \dots + n \text{ terms}) = 2 \times n = 2n$$

Answer: (B)

Question

8. If $1 + 2 + 3 + \dots + n = k$, then $1^3 + 2^3 + 3^3 + \dots + n^3$ is equal to

- (A) k^2
- (B) k^3
- (C) $\frac{k(k+1)}{2}$
- (D) $(k + 1)$

Solution

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} = k$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 = k^2$$

Answer: (A)

Question

9. If an A.P. is $3, 15, 27, 39, \dots, 171$, then the 6th term from the last is

- (A) 99
- (B) 111
- (C) 100
- (D) 171

Solution

Let us consider the A.P in reverse order: 171, 159, 147, ..., 39, 27, 15, 3

Here $a = 171$, $d = t_2 - t_1 = 159 - 171 = -12$

$$\begin{aligned}t_n &= a + (n - 1)d \\n = 6 &\implies t_6 = 171 + (6 - 1)(-12) \\&= 171 + 5(-12) = 171 - 60 = 111\end{aligned}$$

Answer: (B)

Question

10. 3, x , 6.75 are consecutive terms of a G.P then x is

- (A) 2.25
- (B) 9.75
- (C) 4.5
- (D) 1.5

Solution

Given 3, x , 6.75 are in G.P.

$$\begin{aligned}\frac{t_2}{t_1} &= \frac{t_3}{t_2} \\ \frac{x}{3} &= \frac{6.75}{x} \\ x^2 &= 6.75 \times 3 \\ x^2 &= 20.25 \implies x = 4.5\end{aligned}$$

Answer: (C)

Question

11. If 10th term of A.P. is 52 and 16th term of A.P. is 82 then the n^{th} term of this A.P is

- (A) $n + 2$
- (B) $5n - 2$
- (C) $5n + 2$
- (D) $5n$

Solution

$$\begin{aligned}t_{10} &= a + 9d = 52 \quad \dots (1) \\ t_{16} &= a + 15d = 82 \quad \dots (2)\end{aligned}$$

$$(2) - (1) \implies 6d = 30 \implies d = 5$$

$$\text{Sub } d = 5 \text{ in (1): } a + 9(5) = 52 \implies a + 45 = 52 \implies a = 7$$

$$t_n = a + (n - 1)d = 7 + (n - 1)5 = 7 + 5n - 5 = 5n + 2$$

Answer: (C)

Question

12. The quotient and remainder when 21 is divided by -4 are

- (A) 5, -1
- (B) -5, 1
- (C) -5, -1
- (D) 5, 1

Solution

Given $a = 21, b = -4$. By Euclid's lemma $a = bq + r$, where $0 \leq r < |b|$.

$$21 = -4q + r \implies 21 = -4(-5) + 1$$

The quotient $q = -5$ and remainder $r = 1$.

Answer: (B)

Question

13. Find the sum of $1 + 3 + 5 + \dots$ to 40 terms

- (A) 1600
- (B) 1060
- (C) 1640
- (D) 1460

Solution

Sum of first n odd natural numbers is $\sum_{k=1}^n (2k - 1) = n^2$.

Here $n = 40 \implies 1 + 3 + 5 + \dots + 40 \text{ terms} = 40^2 = 1600$.

Answer: (A)

Question

14. The 8th term of the sequence 1, 1, 2, 3, 5, 8, ... is

- (A) 25
- (B) 24
- (C) 23
- (D) 21

Solution

This is a Fibonacci sequence.

1st = 1, 2nd = 1, 3rd = 1+1=2, 4th = 1+2=3, 5th = 2+3=5, 6th = 3+5=8.

7th term = 5 + 8 = 13

8th term = sum of previous two terms = 8 + 13 = 21.

Answer: (D)

Question

15. Using Euclid's division lemma, if any positive integer is divided by 6 the possible remainders are

- (A) zero
- (B) 2,3,6
- (C) 0,1,2,3,4,5
- (D) not defined

Solution

By Euclid's division lemma, $a = bq + r$ where $0 \leq r < b$.

Here $b = 6 \implies 0 \leq r < 6$.

Possible remainders are 0, 1, 2, 3, 4, 5.

Answer: (C)

Question

16. The number of terms of the A.P, 3, 7, 11, 15... to be taken so that the sum is 406 is

- (A) 5
- (B) 10
- (C) 12
- (D) 14

Solution

$a = 3, d = 4, S_n = 406$

$$\frac{n}{2}[2a + (n-1)d] = 406$$

$$\frac{n}{2}[6 + (n-1)4] = 406$$

$$\frac{n}{2}[6 + 4n - 4] = 406$$

$$n[2n + 1] = 406 \implies 2n^2 + n - 406 = 0$$

$$(2n + 29)(n - 14) = 0 \implies n = 14 \text{ (since } n > 0)$$

Answer: (D)

Question

17. The exponent of 2 in the prime factorisation of 144 is

- (A) 4
- (B) 5
- (C) 6
- (D) 3

Solution

Prime factorization of 144:

$$144 = 2 \times 72 = 2^2 \times 36 = 2^3 \times 18 = 2^4 \times 9 = 2^4 \times 3^2$$

The exponent of 2 is 4.

Answer: (A)

Question

18. If n is any natural number then $6^n - 5^n$ always ends with

- (A) 1
- (B) 3
- (C) 5
- (D) 7

Solution

$$6^1 - 5^1 = 1$$

$$6^2 - 5^2 = 36 - 25 = 11$$

$$6^3 - 5^3 = 216 - 125 = 91$$

It always ends with 1.

Answer: (A)

Question

19. If n is a natural number, then $9^{2n} - 4^{2n}$ is always divisible by

- (A) 5
- (B) 13
- (C) both 5 and 13
- (D) None of these

Solution

$9^{2n} - 4^{2n} = (9^2)^n - (4^2)^n$ is of form $a^{2n} - b^{2n}$, which is divisible by $(a - b)$ and $(a + b)$.

$9 - 4 = 5$ and $9 + 4 = 13$.

It is divisible by both 5 and 13.

Answer: (C)

Question

20. If p and q are co-prime numbers, then p^2 and q^2 are

- (A) co-prime
- (B) not co-prime
- (C) even
- (D) odd

Solution

Let 3 and 5 be co-prime. $3^2 = 9$, $5^2 = 25$, and 9 and 25 are co-prime.

Similarly, if p and q are co-prime, p^2 and q^2 share no common prime factors, so they are also co-prime.

Answer: (A)

Question

21. If the LCM of 'a' and 18 is 36 and the HCF of 'a' and 18 is 2, then $a =$

- (A) 2
- (B) 3
- (C) 4
- (D) 1

Solution

Product of two numbers = LCM $\times$ HCF

$$a \times 18 = 36 \times 2$$
$$a = \frac{36 \times 2}{18} \implies a = 4$$

Answer: (C)

Question

22. If $a = 2^3 \times 3$, $b = 2 \times 3 \times 5$, $c = 3^n \times 5$ and $\text{LCM}(a, b, c) = 2^3 \times 3^2 \times 5$, then $n =$

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Solution

$\text{LCM} = 2^3 \times 3^2 \times 5$.

LCM takes the highest power of each prime. a has 3^1 and b has 3^1 .

To get 3^2 in the LCM, c must contribute it. Thus, $n = 2$.

Answer: (B)

Question

23. The largest number which divides 70 and 125, leaving remainders 5 and 8, respectively.

- (A) 13
- (B) 65
- (C) 875
- (D) 2520

Solution

Required number divides $(70 - 5) = 65$ and $(125 - 8) = 117$ completely.

HCF of 65 and 117:

$65 = 13 \times 5$ and $117 = 13 \times 9$. HCF = 13.

Answer: (A)

Question

24. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$; x, y are prime numbers, then HCF (a, b) is

- (A) xy
- (B) xy^2
- (C) x^3y^3
- (D) x^2y^2

Solution

For HCF, take the smallest power of each common prime factor.

$$a = x^3y^2, b = xy^3 \implies \text{HCF}(a, b) = xy^2.$$

Answer: (B)

Question

25. $8^{5k} \equiv \text{_____} \pmod{7}$

- (A) 0
- (B) 4
- (C) 1
- (D) 2

Solution

$$8 \equiv 1 \pmod{7}$$

$$8^5 \equiv 1^5 \pmod{7} \implies 8^5 \equiv 1 \pmod{7}$$

$$8^{5k} \equiv 1^k \pmod{7} \implies 8^{5k} \equiv 1 \pmod{7}$$

Answer: (C)

Question

26. If the sum of p terms of an A.P is q and the sum of q terms is p , then the sum of $p + q$ terms will be

- (A) 0
- (B) $p - q$
- (C) $p + q$
- (D) $-(p + q)$

Solution

Standard property of A.P: If $S_p = q$ and $S_q = p$, then $S_{p+q} = -(p + q)$.

Answer: (D)

Question

27. If the first, second and last term of an A.P are a, b and $2a$ respectively its sum is

- (A) $\frac{ab}{2(b-a)}$
- (B) $\frac{ab}{b-a}$
- (C) $\frac{3ab}{2(b-a)}$
- (D) none of these

Solution

First term = a , second term = $b \implies d = b - a$. Last term $l = 2a$.

$$n = \frac{l - a}{d} + 1 = \frac{2a - a}{b - a} + 1 = \frac{a + b - a}{b - a} = \frac{b}{b - a}$$
$$S_n = \frac{n}{2}(a + l) = \frac{b}{2(b - a)}(a + 2a) = \frac{3ab}{2(b - a)}$$

Answer: (C)

Question

28. If 7th and 13th terms of an A.P be 34 and 64 respectively, then its 18th term is

- (A) 87
- (B) 88
- (C) 89
- (D) 90

Solution

$$t_7 = a + 6d = 34 \quad \dots (1)$$

$$t_{13} = a + 12d = 64 \quad \dots (2)$$

$$(2) - (1) \implies 6d = 30 \implies d = 5.$$

$$\text{Sub in (1): } a + 6(5) = 34 \implies a = 4.$$

$$t_{18} = a + 17d = 4 + 17(5) = 4 + 85 = 89.$$

Answer: (C)

Question

29. The first and last terms of an A.P are 1 and 11. If the sum of its terms is 36, then the number of terms will be

- (A) 5
- (B) 6
- (C) 7
- (D) 8

Solution

$$a = 1, l = 11, S_n = 36.$$

$$S_n = \frac{n}{2}[a + l]$$

$$36 = \frac{n}{2}[1 + 11]$$

$$36 = \frac{n}{2}[12]$$

$$36 = 6n \implies n = 6$$

Answer: (B)

Question

30. If the sum of three consecutive terms of an increasing A.P is 51 and the product of the first and third of these terms is 273, then the third term is

- (A) 13
- (B) 9
- (C) 21
- (D) 17

Solution

Let terms be $a - d, a, a + d$.

Sum = $a - d + a + a + d = 51 \implies 3a = 51 \implies a = 17$.

Product of 1st and 3rd: $(a - d)(a + d) = 273$.

$$17^2 - d^2 = 273$$

$$289 - 273 = d^2 \implies 16 = d^2 \implies d = 4 \text{ (since increasing A.P)}$$

Third term = $a + d = 17 + 4 = 21$.

Answer: (C)

Question

31. If b divides a then

- (A) $r \neq 0$
- (B) $a = bq$
- (C) $a \neq bq$
- (D) $b > a$

Solution

By Euclid's lemma, $a = bq + r$. If b divides a completely, the remainder $r = 0$.

Therefore, $a = bq$.

Answer: (B)

Question

32. If the first term of an A.P is 2 and common difference is 4 then the sum of its 40 terms is

- (A) 3200
- (B) 1600
- (C) 200
- (D) 2800

Solution

Given $a = 2$, $d = 4$, $n = 40$.

$$\begin{aligned}S_n &= \frac{n}{2}[2a + (n - 1)d] \\S_{40} &= \frac{40}{2}[2(2) + (40 - 1)4] \\&= 20[4 + 39(4)] = 20(4 + 156) \\&= 20(160) = 3200\end{aligned}$$

Answer: (A)

Chapter 3: Algebra

Question

1. The G.C.D of a^m, a^{m+1}, a^{m+2} is

- (A) a^m
- (B) a^{m+1}
- (C) a^{m+2}
- (D) 1

Solution

$f(x) = a^m$
 $g(x) = a^{m+1} = a^m \cdot a$
 $h(x) = a^{m+2} = a^m \cdot a^2$
Thus, GCD = a^m
Answer: (A)

Question

2. $\frac{a^2}{a^2-b^2} + \frac{b^2}{b^2-a^2} =$

- (A) $a - b$
- (B) $a + b$
- (C) $a^2 - b^2$
- (D) 1

Solution

$$\begin{aligned}\frac{a^2}{a^2-b^2} + \frac{b^2}{b^2-a^2} &= \frac{a^2}{a^2-b^2} + \frac{b^2}{-(a^2-b^2)} \\&= \frac{a^2}{a^2-b^2} - \frac{b^2}{a^2-b^2} \\&= \frac{a^2-b^2}{a^2-b^2} = 1\end{aligned}$$

Answer: (D)

Question

3. The non-diagonal elements in any unit matrix are

- (A) 0
- (B) 1
- (C) m
- (D) n

Solution

In a unit (identity) matrix, all main diagonal elements are 1, and all non-diagonal elements are 0.

Answer: (A)

Question

4. The LCM of $x^3 - a^3$ and $(x - a)^2$ is

- (A) $(x^3 - a^3)(x + a)$
- (B) $(x^3 - a^3)(x - a)^2$
- (C) $(x - a)^2(x^2 + ax + a^2)$
- (D) $(x + a)^2(x^2 + ax + a^2)$

Solution

$$f(x) = x^3 - a^3 = (x - a)(x^2 + ax + a^2)$$

$$g(x) = (x - a)^2$$

$$\text{Thus, LCM} = (x - a)^2(x^2 + ax + a^2)$$

Answer: (C)

Question

5. The excluded value of the rational expression $\frac{x^3+8}{x^2-2x-8}$ is

- (A) 8
- (B) 2
- (C) 4
- (D) 1

Solution

The expression is undefined when the denominator is zero.

$$x^2 - 2x - 8 = 0$$

$$(x + 2)(x - 4) = 0 \implies x = -2, 4$$

The excluded values are -2, 4. From the options, 4 is correct.

Answer: (C)

Question

6. If a polynomial is a perfect square then its factors will be repeated _____ number of times.

- (A) Odd
- (B) zero
- (C) even
- (D) none of the above

Solution

If a polynomial is a perfect square, each of its factors must occur in pairs (an even number of times).

Answer: (C)

Question

7. The solution of $x^2 - 25 = 0$ is

- (A) No real roots
- (B) real and equal roots
- (C) Real and unequal roots
- (D) imaginary roots

Solution

$$x^2 - 25 = 0 \implies (x + 5)(x - 5) = 0 \implies x = \pm 5$$

The roots are real and unequal.

Answer: (C)

Question

8. For the given matrix $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$ the order of the matrix $(A^T)^T$ is

- (A) 2×3
- (B) 3×2
- (C) 3×4
- (D) 4×3

Solution

Order of the matrix $A = 2 \times 3$

Order of the matrix $A^T = 3 \times 2$

Order of the matrix $(A^T)^T = 2 \times 3$ (since $(A^T)^T = A$)

Answer: (A)

Question

9. On dividing $\frac{x^2-25}{x+3}$ by $\frac{x+5}{x^2-9}$

- (A) $(x - 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 5)(x - 3)$
- (D) $(x + 5)(x + 3)$

Solution

$$\begin{aligned}\frac{x^2 - 25}{x + 3} \div \frac{x + 5}{x^2 - 9} &= \frac{(x + 5)(x - 5)}{x + 3} \times \frac{(x + 3)(x - 3)}{x + 5} \\ &= (x - 5)(x - 3)\end{aligned}$$

Answer: (A)

Question

10. Graph of a quadratic equation is

- (A) Straight line
- (B) Circle
- (C) Parabola
- (D) Hyperbola

Solution

The graph of any quadratic equation of the form $ax^2 + bx + c = 0$ is a parabola.

Answer: (C)

Question

11. Graph of a linear polynomial is a

- (A) Straight line
- (B) Circle
- (C) Parabola
- (D) Hyperbola

Solution

A linear polynomial represents an equation of a straight line on a graph.

Answer: (A)

Question

12. The G.C.D of $6x^2y$, $9x^2yz$, $12x^2y^2z$ is

- (A) $36xy^2z^2$
- (B) $36x^2y^2z$
- (C) $36x^2y^2z^2$
- (D) $3x^2y$

Solution

$$f(x) = 6x^2y = 3 \times 2 \times x^2 \times y$$

$$g(x) = 9x^2yz = 3 \times 3 \times x^2 \times y \times z$$

$$h(x) = 12x^2y^2z = 2 \times 2 \times 3 \times x^2 \times y \times y \times z$$

$$\text{Thus, GCD} = 3 \times x^2 \times y = 3x^2y$$

Answer: (D)

Question

13. The matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ is

- (A) unit matrix
- (B) row matrix
- (C) diagonal matrix
- (D) scalar matrix

Solution

The non-diagonal elements are zero and the diagonal elements are distinct. Therefore, it is a diagonal matrix.

Answer: (C)

Question

14. The sum and product of the roots are 9 and 14 respectively, then their appropriate quadratic equation is

- (A) $x^2 + 9x + 14 = 0$
- (B) $x^2 - 9x + 14 = 0$
- (C) $x^2 - 14x + 9 = 0$
- (D) $x^2 + 14x + 9 = 0$

Solution

Sum of roots = 9 and Product of roots = 14.

$$x^2 - (\text{Sum of roots})x + (\text{Product of roots}) = 0$$

$$x^2 - 9x + 14 = 0$$

Answer: (B)

Question

15. The sum of the roots for the quadratic equation $3\left(x + \frac{2}{3}\right)^2 = 0$ is

- (A) $-\frac{4}{3}$
- (B) -4
- (C) $\frac{4}{3}$
- (D) $\frac{3}{4}$

Solution

$$\begin{aligned} 3\left(x + \frac{2}{3}\right)^2 = 0 &\implies \left(x + \frac{2}{3}\right)^2 = 0 \\ x^2 + 2(x)\left(\frac{2}{3}\right) + \frac{4}{9} &= 0 \\ x^2 + \frac{4}{3}x + \frac{4}{9} &= 0 \end{aligned}$$

$$\text{Sum of the roots} = -\frac{b}{a} = -\frac{4}{3}$$

Answer: (A)

Question

16. The LCM of a^k, a^{k+3}, a^{k+5} where $k \in \mathbb{N}$ is

- (A) a^{k+9}
- (B) a^k
- (C) a^{k+6}
- (D) a^{k+5}

Solution

Let $f(x) = a^k$, $g(x) = a^{k+3} = a^k \cdot a^3$, $h(x) = a^{k+5} = a^k \cdot a^5$.
LCM requires the highest power, which is a^{k+5} .

Answer: (D)

Question

17. If α and β are the zeros of the polynomial $x^2 - 5x + 6$ then $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to

- (A) $-\frac{5}{6}$
- (B) $-\frac{6}{5}$
- (C) $\frac{6}{5}$
- (D) $\frac{5}{6}$

Solution

$x^2 - 5x + 6 = 0$. Compare with $ax^2 + bx + c = 0$. Here $a = 1, b = -5, c = 6$.

$$\alpha + \beta = \frac{-b}{a} = \frac{5}{1} = 5, \quad \alpha\beta = \frac{c}{a} = \frac{6}{1} = 6.$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{5}{6}$$

Answer: (D)

Question

18. If the number of Rows and Columns are equal in matrix, then it is said to be a

- (A) Row matrix
- (B) Rectangular matrix
- (C) Square matrix
- (D) Column matrix

Solution

A matrix with an equal number of rows and columns is a Square matrix.

Answer: (C)

Question

19. Transpose of a row matrix is

- (A) Unit matrix
- (B) diagonal matrix
- (C) column matrix
- (D) row matrix

Solution

The transpose operation swaps rows and columns. Thus, a row matrix becomes a column matrix.

Answer: (C)

Question

20. If $A = \begin{pmatrix} 1 & -2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$ then $A + B^T =$

- (A) $\begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$
- (B) $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$
- (C) $\begin{pmatrix} 2 & 4 & 6 \end{pmatrix}$
- (D) not defined

Solution

$$A = \begin{pmatrix} 1 & -2 & 3 \end{pmatrix}$$

$$B = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} \implies B^T = \begin{pmatrix} -1 & 2 & -3 \end{pmatrix}$$

$$A + B^T = \begin{pmatrix} 1 & -2 & 3 \end{pmatrix} + \begin{pmatrix} -1 & 2 & -3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

Answer: (A)

Question

21. A quadratic equation whose one root is 3 is

- (A) $x^2 - 6x - 5 = 0$
- (B) $x^2 + 6x - 5 = 0$
- (C) $x^2 - 5x - 6 = 0$
- (D) $x^2 - 5x + 6 = 0$

Solution

Testing $x = 3$ on option (D):

$$3^2 - 5(3) + 6 = 9 - 15 + 6 = 0. \text{ Hence, option (D) is correct.}$$

Alternatively, factorize (D): $x^2 - 5x + 6 = (x - 3)(x - 2) = 0 \implies x = 3 \text{ or } x = 2.$

Answer: (D)

Question

22. In solving the system of linear equation in three variables, in any step we get $0 = 0$ then the system has

- (A) no solution
- (B) one solution
- (C) infinitely many solution
- (D) finite solution

Solution

Getting an identity like $0 = 0$ indicates that the equations are dependent, meaning there are infinitely many solutions.

Answer: (C)

Question

23. The HCF of the polynomials $p(x)$ and $q(x)$ is $6x - 9$ then $p(x)$ could be

- (A) $2x - 3$
- (B) $12x - 18, 2$
- (C) $3(2x - 3)^2, 6(2x - 3)$
- (D) $3(2x - 3), 6(2x + 3)$

Solution

$6x - 9 = 3(2x - 3)$. We check every option which has the common term $(6x - 9)$. Option C provides multiples that share this HCF.

Answer: (C)

Question

24. The square root of $x^2 + y^2 + z^2 - 2xy + 2yz - 2zx$ is

- (A) $|x + y - z|$
- (B) $|x - y + z|$
- (C) $|x + y + z|$
- (D) $|x - y - z|$

Solution

$$\begin{aligned}x^2 + (-y)^2 + (-z)^2 + 2x(-y) + 2(-y)(-z) + 2(-z)(x) \\= (x + (-y) + (-z))^2 = (x - y - z)^2\end{aligned}$$

Square root is $\sqrt{(x - y - z)^2} = |x - y - z|$

Answer: (D)

Question

25. If $x^2 + 5kx + 16 = 0$ has no real roots then

- (A) $k > \frac{8}{5}$
- (B) $k > -\frac{8}{5}$
- (C) $-\frac{8}{5} < k < \frac{8}{5}$
- (D) $0 < k < \frac{8}{5}$

Solution

For no real roots, $\Delta < 0$.

$$\begin{aligned}b^2 - 4ac &< 0 \\(5k)^2 - 4(1)(16) &< 0 \\25k^2 - 64 &< 0 \implies k^2 < \frac{64}{25} \implies -\frac{8}{5} < k < \frac{8}{5}\end{aligned}$$

Answer: (C)

Question

26. A polynomial of the form $ax^5 + bx^3 + cx^2 + dx + e$ has at most _____ zeros.

- (A) 3
- (B) 5
- (C) 7
- (D) 11

Solution

The degree of the polynomial is 5. Therefore, it has at most 5 zeros.

Answer: (B)

Question

27. The roots of the quadratic equation $x^2 + 5x - (\alpha + 1)(\alpha + 6) = 0$, where α is a constant are

- (A) $(\alpha + 1), (\alpha + 6)$
- (B) $(\alpha + 1), -(\alpha + 6)$
- (C) $-(\alpha + 1), (\alpha + 6)$
- (D) $-(\alpha + 1), -(\alpha + 6)$

Solution

$$x^2 + 5x - (\alpha + 1)(\alpha + 6) = 0$$

Product of roots $= -(\alpha + 1)(\alpha + 6)$.

Sum of roots $= -5$.

Notice that $(\alpha + 1) - (\alpha + 6) = -5$.

Therefore, the roots are $(\alpha + 1)$ and $-(\alpha + 6)$.

Answer: (B)

Question

28. If the quadratic equation $mx^2 + 2x + m = 0$ has two equal roots, then the values of m are

- (A) +1
- (B) 0, 2
- (C) 0, 1
- (D) -1, 0

Solution

$a = m, b = 2, c = m$. For equal roots, $b^2 - 4ac = 0$.

$$(2)^2 - 4(m)(m) = 0$$

$$4 - 4m^2 = 0$$

$$4m^2 = 4 \implies m^2 = 1 \implies m = \pm 1$$

Answer: (A) (Note: The option only explicitly shows +1, but standardly implies ± 1 per solution).

Question

29. The type of matrix $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ is

- (A) Scalar
- (B) Unit
- (C) Diagonal
- (D) null matrix

Solution

The non-diagonal elements are zero, and the diagonal elements (2 and 3) are different. Thus, it is a Diagonal matrix.

Answer: (C)

Question

30. Addition of matrices is

- (A) not commutative
- (B) not associative
- (C) distributive
- (D) commutative

Solution

For any two matrices A and B of the same order, $A + B = B + A$. Matrix addition is commutative.

Answer: (D)

Question

31. $A = \begin{bmatrix} 2 & 3 \\ -9 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 5 \\ 7 & -1 \end{bmatrix}$. Find the additive inverse of A.

- (A) $\begin{bmatrix} -1 & 2 \\ 16 & -6 \end{bmatrix}$
- (B) $\begin{bmatrix} 1 & -2 \\ -16 & 6 \end{bmatrix}$
- (C) $\begin{bmatrix} 1 & -16 \\ -2 & 6 \end{bmatrix}$
- (D) $\begin{bmatrix} -1 & -2 \\ -16 & -6 \end{bmatrix}$

Solution

$$A = \begin{bmatrix} 2 & 3 \\ -9 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 5 \\ 7 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -16 & 6 \end{bmatrix}$$

Additive inverse is $-A = \begin{bmatrix} -1 & 2 \\ 16 & -6 \end{bmatrix}$

Answer: (A)

Question

32. $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 1 \\ -4 & 9 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 5 \\ 1 & 6 \\ 4 & 7 \end{bmatrix}$. Which of the following exists?

(i) AB exists (ii) BC exists (iii) $AC + B$ exists (iv) AC exists

- (A) (i)(ii)
(B) (iii)(iv)
(C) (iv)
(D) (iii)

Solution

Order of $A = 2 \times 3$, Order of $B = 2 \times 2$, Order of $C = 3 \times 2$.

- AB : cols in A (3) $\neq$ rows in B (2). Not exists.

- BC : cols in B (2) $\neq$ rows in C (3). Not exists.

- AC : cols in A (3) = rows in C (3). Exists (Resulting order 2×2).

- $AC + B$: Since AC is 2×2 and B is 2×2 , addition exists.

Both (iii) and (iv) exist.

Answer: (B)

Question

33. If $b = a + c$ then the equation $ax^2 + bx + c = 0$ has

- (A) real roots
(B) no roots
(C) equal roots
(D) no real roots

Solution

$$\begin{aligned} \Delta &= b^2 - 4ac = (a + c)^2 - 4ac \\ &= a^2 + c^2 + 2ac - 4ac = a^2 + c^2 - 2ac = (a - c)^2 > 0 \end{aligned}$$

Since $\Delta > 0$, the equation has real roots.

Answer: (A)

Chapter 4: Geometry

Question

1. If two circles touch internally, the distance between their centres is

- (A) $r_1 + r_2$
- (B) $r_1 = r_2$
- (C) $r_1 - r_2$
- (D) $r_1 > r_2$

Solution

Let two circles with centres at O and P touch each other at Q.

$OQ = r_1$, $PQ = r_2$ and let $r_1 > r_2$.

Distance between centres $d = OP = OQ - PQ = r_1 - r_2$.

Answer: (C)

Question

2. PQ is a tangent to a circle with centre O at the point P. If $\triangle OPQ$ is an isosceles triangle then $\angle OQP$ is equal to

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

Solution

Tangent is always $\perp$ to the radius at the point of contact. Now, $\angle OPQ = 90^\circ$.

Since $\triangle OPQ$ is isosceles and right-angled, $\angle PQO = \angle POQ$.

$$\angle OPQ + \angle PQO + \angle POQ = 180^\circ$$

$$90^\circ + 2\angle PQO = 180^\circ$$

$$2\angle PQO = 90^\circ \implies \angle PQO = 45^\circ$$

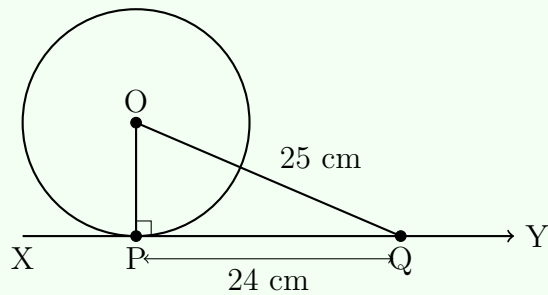
Answer: (B)

Question

3. From a point Q, the length of the tangent to a circle is 24cm and the distance of Q from the centre is 25cm. The radius of the circle is

- (A) 7cm
- (B) 12cm
- (C) 15cm
- (D) 24.5cm

Solution



In right $\triangle OPQ$:

$$OQ^2 = OP^2 + PQ^2$$

$$(25)^2 = OP^2 + (24)^2$$

$$OP^2 = 625 - 576 = 49 \implies OP = \sqrt{49} = 7 \text{ cm}$$

Answer: (A)

Question

4. If TP and TQ are two tangents to a circle with centre O so that $\angle POQ = 110^\circ$ then $\angle PTQ$ is equal to

- (A) 60°
- (B) 70°
- (C) 80°
- (D) 90°

Solution

In quadrilateral TPOQ, $\angle TPO = 90^\circ$ and $\angle TQO = 90^\circ$.

$$\angle TPO + \angle OQT + \angle PTQ + \angle POQ = 360^\circ$$

$$90^\circ + 90^\circ + \angle PTQ + 110^\circ = 360^\circ$$

$$290^\circ + \angle PTQ = 360^\circ$$

$$\angle PTQ = 360^\circ - 290^\circ = 70^\circ$$

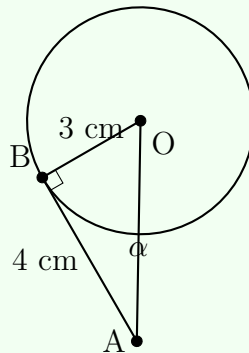
Answer: (B)

Question

5. The length of the tangent from a point A to a circle, of radius 3cm, is 4cm. The distance of A from the centre of the circle is

- (A) $\sqrt{7}$ cm
- (B) 7 cm
- (C) 5 cm
- (D) 25 cm

Solution



$OB = 3 \text{ cm} = r$, $AB = 4 \text{ cm}$.

In $\triangle AOB$, $\angle B = 90^\circ$.

$$AO^2 = AB^2 + BO^2$$

$$\alpha^2 = 16 + 9 = 25 \implies \alpha = \sqrt{25} = 5 \text{ cm}$$

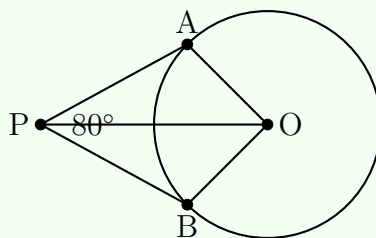
Answer: (C)

Question

6. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° then $\angle POA$ is equal to

- (A) 50°
- (B) 60°
- (C) 70°
- (D) 80°

Solution



$\triangle OAP \cong \triangle OBP$ (SSS congruency).

So, $\angle OPA = \angle OPB$.

$$\angle OPA = \frac{1}{2} \angle APB = \frac{1}{2} \times 80^\circ = 40^\circ.$$

In $\triangle OPA$, $\angle POA + \angle OPA + \angle OAP = 180^\circ$.

$$\angle POA + 40^\circ + 90^\circ = 180^\circ$$

$$\angle POA = 180^\circ - 130^\circ = 50^\circ$$

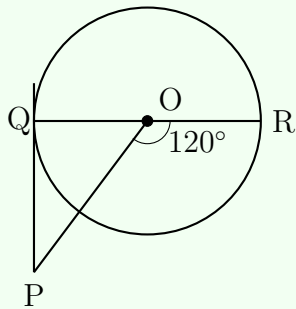
Answer: (A)

Question

7. PQ is a tangent drawn from a point P to a circle with centre O and OQR is a diameter of the circle such that $\angle POR = 120^\circ$ then $\angle OPQ$ is

- (A) 60°
- (B) 45°
- (C) 30°
- (D) 90°

Solution



$$\angle OQP = 90^\circ.$$

$$\angle QOP = 180^\circ - 120^\circ = 60^\circ \text{ (Linear pair).}$$

$$\angle OPQ = 180^\circ - \angle OQP - \angle QOP = 180^\circ - 90^\circ - 60^\circ = 30^\circ.$$

Answer: (C)

Question

8. The length of the tangent drawn from a point 8cm away from the centre of a circle of radius 6cm is

- (A) $\sqrt{7}$ cm
- (B) $2\sqrt{7}$ cm
- (C) 10 cm
- (D) 5 cm

Solution

Radius = 6 cm and hypotenuse $OT = 8$ cm.

Let the tangent be PT .

$$\begin{aligned} PT &= \sqrt{8^2 - 6^2} \\ &= \sqrt{64 - 36} = \sqrt{28} = 2\sqrt{7} \text{ cm} \end{aligned}$$

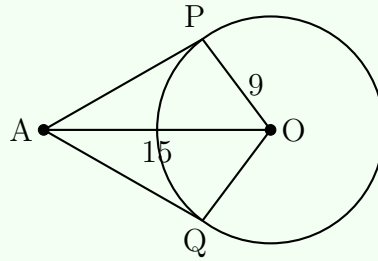
Answer: (B)

Question

9. AP and PQ are tangents drawn from a point A to a circle with centre O and radius 9cm. If $OA = 15\text{cm}$, then $AP + AQ =$

- (A) 12cm
- (B) 18cm
- (C) 24cm
- (D) 36cm

Solution



Tangents drawn from an external point will be equal in length. So, $AP = AQ$.

In right triangle APO:

$$AP = \sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144} = 12 \text{ cm.}$$

$$AQ = 12 \text{ cm.}$$

$$AP + AQ = 12 + 12 = 24 \text{ cm.}$$

Answer: (C)

Question

10. The areas of two similar triangles are respectively 9cm^2 and 16cm^2 . The ratio of their corresponding sides is

- (A) 3:4
- (B) 4:3
- (C) 2:3
- (D) 4:5

Solution

$$\frac{\text{Area}(\text{Triangle 1})}{\text{Area}(\text{Triangle 2})} = \left(\frac{\text{Side 1}}{\text{Side 2}}\right)^2$$
$$\frac{9}{16} = \left(\frac{\text{Side 1}}{\text{Side 2}}\right)^2$$

Taking square root on both sides: $\frac{\text{Side 1}}{\text{Side 2}} = \frac{3}{4}$.

The ratio of their corresponding sides is 3:4.

Answer: (A)

Question

11. If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{2}{5}$ then $\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle DEF)} =$

- (A) 2:5
- (B) 4:25
- (C) 4:15
- (D) 8:125

Solution

Since $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{2}{5}$, $\triangle ABC \sim \triangle DEF$.

$$\begin{aligned}\frac{\text{Area } \triangle ABC}{\text{Area } \triangle DEF} &= \left(\frac{AB}{DE}\right)^2 \\ &= \left(\frac{2}{5}\right)^2 = \frac{4}{25}\end{aligned}$$

Ratio is 4:25.

Answer: (B)

Question

12. If $\triangle ABC$ and $\triangle DEF$ are similar such that $2AB = DE$ and $BC = 8\text{cm}$, then $EF =$

- (A) 16 cm
- (B) 12 cm
- (C) 8 cm
- (D) 4 cm

Solution

$\triangle ABC \sim \triangle DEF$.

$DE = 2AB$, $BC = 8\text{ cm}$.

By the property of similar triangles:

$$\begin{aligned}\frac{AB}{DE} &= \frac{BC}{EF} \\ \frac{AB}{2AB} &= \frac{8}{EF} \\ \frac{1}{2} &= \frac{8}{EF} \implies EF = 16\text{ cm}\end{aligned}$$

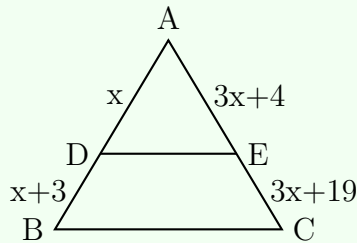
Answer: (A)

Question

13. In figure, the value of x for which $DE \parallel BC$ is

- (A) 4
- (B) 1
- (C) 3
- (D) 2

Solution



By Thales theorem $\frac{AD}{DB} = \frac{AE}{EC}$.
 $AD \times EC = AE \times DB$

$$\begin{aligned}(x)(3x + 19) &= (3x + 4)(x + 3) \\ 3x^2 + 19x &= 3x^2 + 9x + 4x + 12 \\ 19x &= 13x + 12 \\ 19x - 13x &= 12 \\ 6x &= 12 \implies x = 2\end{aligned}$$

Answer: (D)

Chapter 5: Coordinate Geometry

Question

1. The equation of a line passing through the origin and parallel to the line $7x - 3y + 4 = 0$ is

- (A) $7x - 3y + 4 = 0$
- (B) $3x - 7y + 4 = 0$
- (C) $3x + 7y = 0$
- (D) $7x - 3y = 0$

Solution

line passing through the origin $(0, 0)$

It is form of $y = mx + c$, $7x - 3y + 4 = 0$

$$-3y = -7x - 4 = 0 \implies y = \frac{7}{3}x + \frac{4}{3}$$

Slope $m = \frac{7}{3}$

Slope-point form of the parallel line: $(y - y_1) = m(x - x_1)$

$$y - 0 = \frac{7}{3}(x - 0) \implies 3y = 7x \implies 7x - 3y = 0$$

Answer: (D) $7x - 3y = 0$

Question

2. The slope of the straight line perpendicular to x-axis is

- (A) 1
- (B) 0
- (C) ∞
- (D) -1

Solution

The slope of the x-axis is zero.

Since, the slope of the perpendicular line $\implies -\frac{1}{\text{slope of the given line}}$

$$\implies -\frac{1}{0} = \infty$$

Answer: (C) ∞

Question

3. The perimeter of a triangle formed by the points $(0, 0)$, $(1, 0)$ and $(0, 1)$ is

- (A) $\sqrt{2}$
- (B) 2
- (C) $2 + \sqrt{2}$
- (D) $2 - \sqrt{2}$

Solution

$$AB = \sqrt{(1-0)^2 + (0-0)^2} = \sqrt{1} = 1$$

$$BC = \sqrt{(0-1)^2 + (1-0)^2} = \sqrt{2}$$

$$AC = \sqrt{(0-0)^2 + (1-0)^2} = \sqrt{1} = 1$$

$$AB + BC + AC = 1 + \sqrt{2} + 1 = 2 + \sqrt{2} \text{ units}$$

Answer: (C) $2 + \sqrt{2}$

Question

4. If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$ and $D(p, 3)$ are the vertices of a parallelogram, taken in order then the value of p is

- (A) -7
- (B) 7
- (C) 6
- (D) -6

Solution

Midpoint of diagonal AC = Midpoint of diagonal BD

$$\left(\frac{6+9}{2}, \frac{1+4}{2}\right) = \left(\frac{8+p}{2}, \frac{3+2}{2}\right)$$

$$\left(\frac{15}{2}, \frac{5}{2}\right) = \left(\frac{8+p}{2}, \frac{5}{2}\right)$$

Equating x coordinates:

$$\frac{15}{2} = \frac{8+p}{2} \implies 15 = 8 + p \implies p = 7$$

Answer: (B) 7

Question

5. The condition of two straight lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ where the co-efficients are non-zero, are perpendicular if and only if

- (A) $a_1b_2 - a_2b_1 = 0$
- (B) $a_1b_2 + a_2b_1 = 0$
- (C) $a_1a_2 + b_1b_2 = 0$
- (D) $a_1a_2 - b_1b_2 = 0$

Solution

Condition for perpendicular lines is $m_1 \times m_2 = -1$.

$$\left(-\frac{a_1}{b_1}\right) \times \left(-\frac{a_2}{b_2}\right) = -1 \implies a_1 a_2 = -b_1 b_2 \implies a_1 a_2 + b_1 b_2 = 0$$

Answer: (C) $a_1 a_2 + b_1 b_2 = 0$

Question

6. Find the equation of the straight line passing through (5, 7) and is parallel to X axis

- (A) $X = 0$
- (B) $X = 5$
- (C) $Y = 7$
- (D) $Y = 5$

Solution

The equation of the line passing through the point (5, 7) which is parallel to the X axis is $Y = 7$.

Answer: (C) $Y = 7$

Question

7. The fourth vertex of a parallelogram ABCD whose three vertices are $A(-2, 5)$, $B(6, 7)$, $C(8, 3)$ is

- (A) (0, 1)
- (B) (0, -1)
- (C) (-1, 0)
- (D) (1, 0)

Solution

The fourth vertex of a parallelogram is $D(x_4, y_4)$.

$$\text{Midpoints of } AC = \left(\frac{-2+8}{2}, \frac{5+3}{2}\right) = \left(\frac{6}{2}, \frac{8}{2}\right) = (3, 4)$$

$$\text{Midpoints of } BD = \left(\frac{6+x_4}{2}, \frac{7+y_4}{2}\right)$$

Midpoints of AC = Midpoints of BD

$$\left(\frac{6+x_4}{2}, \frac{7+y_4}{2}\right) = (3, 4)$$

$$\frac{6+x_4}{2} = 3 \implies 6 + x_4 = 6 \implies x_4 = 0$$

$$\frac{7+y_4}{2} = 4 \implies 7 + y_4 = 8 \implies y_4 = 1$$

$$(x_4, y_4) = (0, 1)$$

Answer: (A) (0, 1)

Question

8. The area of triangle formed by the points $(-4, 0)$, $(0, -4)$ and $(4, 0)$ is

- (A) 0 sq. Units
- (B) 16 sq. Units
- (C) 4 sq. Units
- (D) none of these

Solution

Area of the triangle:

$$\begin{aligned} &= \frac{1}{2} |(-4)(-4) + (0)(0) + (4)(0) - ((0)(0) + (-4)(4) + (0)(-4))| \\ &= \frac{1}{2} [16 - (-16)] \\ &= \frac{1}{2} (32) = 16 \text{ sq. units} \end{aligned}$$

Answer: (B) 16 sq. Units

Question

9. The area of quadrilateral formed by the points $(-1, 1)$, $(1, 1)$, $(1, -1)$ and $(-1, -1)$ is

- (A) 0 sq.units
- (B) 4 sq.units
- (C) 25 sq.units
- (D) 1 sq.units

Solution

$A(-1, -1), B(1, -1), C(1, 1), D(-1, 1)$

$$A = \frac{1}{2} \{(-1)(-1) + (1)(1) + (1)(1) + (-1)(-1) - ((-1)(1) + (-1)(1) + (1)(-1) + (1)(-1))\}$$

$$A = \frac{1}{2} \{(1 + 1 + 1 + 1) - (-1 - 1 - 1 - 1)\}$$

$$A = \frac{1}{2} \{(4) - (-4)\} = \frac{1}{2} \{8\} = 4$$

Answer: (B) 4 sq.units

Question

10. The point of intersection of the straight lines $y = 0$ and $x = -2$ is

- (A) $(0, -2)$
- (B) $(-2, 0)$
- (C) $(0, 4)$
- (D) $(2, 0)$

Solution

By plotting $y = 0$ and $x = -2$ on a graph.

We can observe that $(-2, 0)$ is a point of intersection.

Answer: (B) $(-2, 0)$

Question

11. The inclination of X axis and every line parallel to X axis is

- (A) 0°
- (B) 90°
- (C) 45°
- (D) 60°

Solution

The inclination of X axis and every line parallel to X axis is 0° .

Answer: (A) 0°

Question

12. The distance of the point $P(-6, -8)$ from the origin is

- (A) 10
- (B) $5\sqrt{2}$
- (C) $\sqrt{10}$
- (D) 100

Solution

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$(x_1, y_1) = (0, 0) \text{ and } (x_2, y_2) = (-6, -8)$$

$$\begin{aligned} d &= \sqrt{(-6 - 0)^2 + (-8 - 0)^2} \\ &= \sqrt{(-6)^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10 \end{aligned}$$

Answer: (A) 10

Question

13. The distance between the points $(a \cos 25^\circ, 0)$ and $(0, a \cos 65^\circ)$ is

- (A) a
- (B) 2a
- (C) 3a
- (D) None of these

Solution

$$A(a \cos 25^\circ, 0) = A(a \cos 25^\circ, 0)$$

$$B(0, a \cos 65^\circ) = B(0, a \sin 25^\circ)$$

$$\begin{aligned} AB^2 &= (a \cos 25^\circ - 0)^2 + (0 - a \sin 25^\circ)^2 \\ &= a^2 \cos^2 25^\circ + a^2 \sin^2 25^\circ \\ &= a^2 (\cos^2 25^\circ + \sin^2 25^\circ) \\ AB^2 &= a^2 (1) \implies AB = \sqrt{a^2} = a \end{aligned}$$

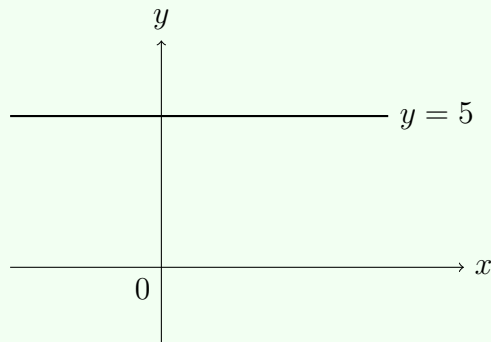
Answer: (A) a

Question

14. The straight line given by the equation $y = 5$ is

- (A) Parallel to X axis
- (B) parallel to Y axis
- (C) passing through the origin
- (D) passing through the point $(2, -5)$

Solution



Parallel to X axis.

Answer: (A) Parallel to X axis

Question

15. If the distance between the point $(4, p)$ and $(1, 0)$ is 5, then $p =$

- (A) 5
- (B) 4
- (C) -4
- (D) 0

Solution

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{(1 - 4)^2 + (0 - p)^2} = 5$$

Square on both sides:

$$(-3)^2 + p^2 = 25$$

$$p^2 = 16 \implies p = \pm 4$$

Answer: ± 4 (Closest option given context is (B) 4 and (C) -4, assuming absolute value or typo in original options)

Question

16. If $(x, 2)$, $(-3, -4)$ and $(7, -5)$ are collinear, then $x =$

- (A) 60
- (B) 63
- (C) -63
- (D) -60

Solution

$$\frac{1}{2}\{x(-4 - (-5)) + (-3)(-5 - 2) + 7(2 - (-4))\} = 0$$

$$\frac{1}{2}\{x(1) + (-3)(-7) + 7(6)\} = 0$$

$$x + 21 + 42 = 0$$

$$x + 63 = 0 \implies x = -63$$

Answer: (C) -63

Question

17. If the area of the triangle formed by the points $(x, 2x)$, $(-2, 6)$ and $(3, 1)$ is 5 sq.units. then $x =$

- (A) $\frac{2}{3}$
- (B) $\frac{5}{3}$
- (C) 3
- (D) 5

Solution

Given $(x, 2x)$, $(3, 1)$, $(-2, 6)$

$$\frac{1}{2}\{x(1 - 6) + 3(6 - 2x) + (-2)(2x - 1)\} = 5 \text{ (Note: using absolute values/order correctly)}$$

The provided text simplifies to:

$$x + 18 - 4x - 6x + 2 - 6x = 10$$

$$20 - 15x = 10 \implies -15x = -10 \implies x = \frac{2}{3}$$

Answer: (A) $\frac{2}{3}$

Question

18. The equations of the sides AB, BC and CA of $\triangle ABC$ are $y - x = 2$, $x + 2y = 1$ and $3x + y + 5 = 0$ respectively. The equation of the altitude through B is

- (A) $x - 3y + 1 = 0$
- (B) $x - 3y + 4 = 0$
- (C) $3x - y + 2 = 0$
- (D) none of these

Solution

The point of intersection of AB and BC is B.

$$y - x = 2 \quad \dots (1)$$

$$2y + x = 1 \quad \dots (2)$$

$$\text{Adding (1) and (2): } 3y = 3 \implies y = 1$$

$$\text{Sub in (1): } 1 - x = 2 \implies x = -1 \implies B(-1, 1)$$

Altitude through B is $AC \perp BD$.

The equation of line perpendicular to $3x + y + 5 = 0$ is $x - 3y + k = 0$.

$$(-1, 1) \text{ passes through } x - 3y + k = 0 \implies -1 - 3(1) + k = 0 \implies k = 4.$$

The equation is $x - 3y + 4 = 0$.

Answer: (B) $x - 3y + 4 = 0$

Chapter 6: Trigonometry

Question

1. If $\tan \theta + \cot \theta = 2$ then $\tan^2 \theta + \cot^2 \theta$ is equal to

- (A) 0
- (B) 1
- (C) 2
- (D) 4

Solution

Given $\tan \theta + \cot \theta = 2$

Squaring on both sides, $(\tan \theta + \cot \theta)^2 = 2^2$

$$\tan^2 \theta + \cot^2 \theta + 2 \tan \theta \cot \theta = 4$$

$$\tan^2 \theta + \cot^2 \theta + 2 \tan \theta \left(\frac{1}{\tan \theta} \right) = 4$$

$$\tan^2 \theta + \cot^2 \theta + 2(1) = 4$$

$$\tan^2 \theta + \cot^2 \theta = 4 - 2 = 2$$

Answer: (C) 2

Question

2. The angle of elevation and depression are usually measured by a device called

- (A) Clinometer
- (B) kaleidoscope
- (C) Periscope
- (D) Telescope

Solution

Answer: (A) Clinometer

Question

3. $\frac{\sin(90^\circ - \theta) \sin \theta}{\tan \theta} + \frac{\cos(90^\circ - \theta) \cos \theta}{\cot \theta} =$

- (A) $\tan \theta$
- (B) 1
- (C) -1
- (D) $\sin \theta$

Solution

$$\begin{aligned} &= \frac{\cos \theta \sin \theta}{\frac{\sin \theta}{\cos \theta}} + \frac{\sin \theta \cos \theta}{\frac{\cos \theta}{\sin \theta}} \\ &= \cos^2 \theta + \sin^2 \theta = 1 \end{aligned}$$

Answer: (B) 1

Question

4. If the line of sight and angle of elevation is given, then which trigonometric ratio is used to find the height of the building?

- (A) $\sin \theta$
- (B) $\cos \theta$
- (C) $\tan \theta$
- (D) $\csc \theta$

Solution

Answer: (C) $\tan \theta$

Question

5. $(1 - \sin^2 \theta) \sec^2 \theta =$

- (A) 0
- (B) 1
- (C) $\tan^2 \theta$
- (D) $\cos^2 \theta$

Solution

$$(1 - \sin^2 \theta) \sec^2 \theta = \cos^2 \theta \sec^2 \theta = 1$$

Answer: (B) 1

Question

6. $(1 + \tan^2 \theta) \sin^2 \theta =$

- (A) $\sin^2 \theta$
- (B) $\cos^2 \theta$
- (C) $\tan^2 \theta$
- (D) $\cot^2 \theta$

Solution

$$(1 + \tan^2 \theta) \sin^2 \theta = \sec^2 \theta \sin^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$$

Answer: (C) $\tan^2 \theta$

Question

7. If $\tan \theta = \frac{a}{x}$ then the value of $\frac{x}{\sqrt{a^2 + x^2}} =$

- (A) $\cos \theta$
- (B) $\sin \theta$
- (C) $\csc \theta$
- (D) $\sec \theta$

Solution

$$\begin{aligned} \frac{x}{\sqrt{a^2 + x^2}} &= \frac{1}{\sqrt{\frac{a^2}{x^2} + 1}} \quad (\text{divide by } x) \\ &= \frac{1}{\sqrt{\tan^2 \theta + 1}} = \frac{1}{\sqrt{\sec^2 \theta}} = \frac{1}{\sec \theta} = \cos \theta \end{aligned}$$

Answer: (A) $\cos \theta$

Question

8. If $x = a \sec \theta$, $y = b \tan \theta$ then the value of $\frac{x^2}{a^2} - \frac{y^2}{b^2} =$

- (A) 1
- (B) -1
- (C) $\tan^2 \theta$
- (D) $\csc^2 \theta$

Solution

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{a^2 \sec^2 \theta}{a^2} - \frac{b^2 \tan^2 \theta}{b^2} = \sec^2 \theta - \tan^2 \theta = 1$$

Answer: (A) 1

Question

9. $\frac{\sec \theta}{\cot \theta + \tan \theta} =$

- (A) $\cot \theta$
- (B) $\tan \theta$
- (C) $\sin \theta$
- (D) $\cot \theta$

Solution

$$\begin{aligned} \frac{\sec \theta}{\cot \theta + \tan \theta} &= \frac{\frac{1}{\cos \theta}}{\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta}} \\ &= \frac{\frac{1}{\cos \theta}}{\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta}} \\ &= \frac{\frac{1}{\cos \theta}}{\frac{1}{\sin \theta \cos \theta}} = \frac{1}{\cos \theta} \times \frac{\sin \theta \cos \theta}{1} = \sin \theta \end{aligned}$$

Answer: (C) $\sin \theta$

Question

10. $(1 - \cos^2 \theta)(1 + \cot^2 \theta) =$

- (A) $\sin^2 \theta$
- (B) 0
- (C) 1
- (D) $\tan^2 \theta$

Solution

$$(1 - \cos^2 \theta)(1 + \cot^2 \theta) = \sin^2 \theta \csc^2 \theta = 1$$

Answer: (C) 1

Question

11. $1 - \frac{\sin^2 \theta}{1 + \cos \theta} =$

- (A) $\cos \theta$
- (B) $\tan \theta$
- (C) $\cot \theta$
- (D) $\csc \theta$

Solution

$$\begin{aligned} 1 - \frac{\sin^2 \theta}{1 + \cos \theta} &= \frac{(1 + \cos \theta) - (1 - \cos^2 \theta)}{1 + \cos \theta} \\ &= \frac{1 + \cos \theta - 1 + \cos^2 \theta}{1 + \cos \theta} \\ &= \frac{\cos \theta(1 + \cos \theta)}{1 + \cos \theta} = \cos \theta \end{aligned}$$

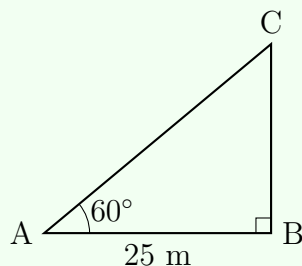
Answer: (A) $\cos \theta$

Question

12. In the adjoining figure, $AC =$

- (A) 25 m
- (B) $25\sqrt{3}$ m
- (C) $\frac{25}{\sqrt{3}}$ m
- (D) $25\sqrt{2}$ m

Solution



$$\tan 60^\circ = \frac{AC}{25} \implies AC = 25 \tan 60^\circ = 25\sqrt{3} \text{ m.}$$

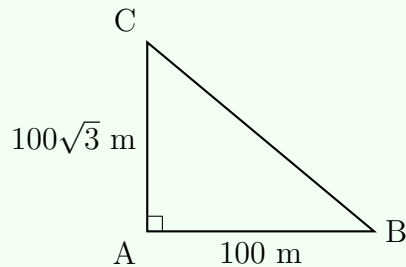
Answer: (B) $25\sqrt{3}$ m

Question

13. In the adjoining figure $\angle ABC =$

- (A) 45°
- (B) 30°
- (C) 60°
- (D) 50°

Solution



$$\tan \angle ABC = \frac{100\sqrt{3}}{100} \implies \tan \angle ABC = \sqrt{3} \implies \angle ABC = 60^\circ$$

Answer: (C) 60°

Question

14. $\cos^4 x - \sin^4 x =$

- (A) $2 \sin^2 x - 1$
- (B) $2 \cos^2 x - 1$
- (C) $1 + 2 \sin^2 x$
- (D) $1 - 2 \cos^2 x$

Solution

$$\begin{aligned}\cos^4 x - \sin^4 x &= (\cos^2 x + \sin^2 x)(\cos^2 x - \sin^2 x) \\ &= \cos^2 x - \sin^2 x \\ &= \cos^2 x - (1 - \cos^2 x) = 2 \cos^2 x - 1\end{aligned}$$

Answer: (B) $2 \cos^2 x - 1$

Question

15. $\sin(90^\circ - \theta) \cos \theta + \cos(90^\circ - \theta) \sin \theta =$

- (A) 1
- (B) -1
- (C) 2
- (D) 0

Solution

$$\begin{aligned}\sin(90^\circ - \theta) \cos \theta + \cos(90^\circ - \theta) \sin \theta \\ = \cos \theta \cos \theta + \sin \theta \sin \theta = \cos^2 \theta + \sin^2 \theta = 1\end{aligned}$$

Answer: (A) 1

Question

16. $(1 + \tan^2 \theta)(1 - \sin \theta)(1 + \sin \theta) =$

- (A) $\cos^2 \theta - \sin^2 \theta$
- (B) $\sin^2 \theta - \cos^2 \theta$
- (C) $\sin^2 \theta + \cos^2 \theta$
- (D) 0

Solution

$$\begin{aligned}(1 + \tan^2 \theta)(1 - \sin \theta)(1 + \sin \theta) &= (1 + \tan^2 \theta)(1 - \sin^2 \theta) \\ &= \sec^2 \theta \cdot \cos^2 \theta \\ &= \frac{1}{\cos^2 \theta} \cdot \cos^2 \theta = 1 \\ &= \sin^2 \theta + \cos^2 \theta\end{aligned}$$

Answer: (C) $\sin^2 \theta + \cos^2 \theta$

Question

17. $(1 + \cot^2 \theta)(1 - \cos \theta)(1 + \cos \theta) =$

- (A) $\tan^2 \theta - \sec^2 \theta$
- (B) $\sin^2 \theta - \cos^2 \theta$
- (C) $\sec^2 \theta - \tan^2 \theta$
- (D) $\cos^2 \theta - \sin^2 \theta$

Solution

$$\begin{aligned}(1 + \cot^2 \theta)(1 - \cos \theta)(1 + \cos \theta) &= \csc^2 \theta \sin^2 \theta \\ &= \frac{1}{\sin^2 \theta} \cdot \sin^2 \theta = 1 \\ &= \sec^2 \theta - \tan^2 \theta\end{aligned}$$

Answer: (C) $\sec^2 \theta - \tan^2 \theta$

Question

18. $(\cos^2 \theta - 1)(\cot^2 \theta + 1) + 1 =$

- (A) 1
- (B) 0
- (C) 2
- (D) -1

Solution

$$\begin{aligned}(\cos^2 \theta - 1)(\cot^2 \theta + 1) + 1 &= (-\sin^2 \theta \times \csc^2 \theta) + 1 \\ &= -1 + 1 = 0\end{aligned}$$

Answer: (B) 0

Question

19. $\frac{1+\tan^2 \theta}{1+\cot^2 \theta} =$

- (A) $\cos^2 \theta$
- (B) $\tan^2 \theta$
- (C) $\sin^2 \theta$
- (D) $\cot^2 \theta$

Solution

$$\frac{1+\tan^2 \theta}{1+\cot^2 \theta} = \frac{\sec^2 \theta}{\csc^2 \theta} = \frac{\frac{1}{\cos^2 \theta}}{\frac{1}{\sin^2 \theta}} = \frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$$

Answer: (B) $\tan^2 \theta$

Question

20. $\sin^2 \theta - \frac{1}{1+\tan^2 \theta} =$

- (A) $\csc^2 \theta + \cot^2 \theta$
- (B) $\csc^2 \theta - \cot^2 \theta$
- (C) $\cot^2 \theta - \csc^2 \theta$
- (D) $\sin^2 \theta - \cos^2 \theta$

Solution

$$\sin^2 \theta - \frac{1}{1+\tan^2 \theta} = \sin^2 \theta - \frac{1}{\sec^2 \theta} = \sin^2 \theta - \cos^2 \theta$$

Answer: (D) $\sin^2 \theta - \cos^2 \theta$

Question

21. $9 \tan^2 \theta - 9 \sec^2 \theta =$

- (A) 1
- (B) 0
- (C) 9
- (D) -9

Solution

$$9 \tan^2 \theta - 9 \sec^2 \theta = -9(\sec^2 \theta - \tan^2 \theta) = -9(1) = -9$$

Answer: (D) -9

Chapter 7: Mensuration

Question

1. The curved surface area of a right circular Cylinder of height 4 cm and base diameter 10 cm is

- (A) 40π sq. units
- (B) 20π sq. units
- (C) 14π sq. units
- (D) 80π sq. units

Solution

Radius of the cylinder = 5 cm, Height of the cylinder = 4 cm

Curved surface area = $2\pi rh$ sq. units = $2\pi(5)(4) = 40\pi$ sq. units

Answer: (A) 40π sq. units

Question

2. If the radius of the cylinder is doubled, the new volume of the cylinder will be _____ times the original volume

- (A) Same
- (B) 3
- (C) 4
- (D) 2

Solution

Volume of the cylinder = $\pi r^2 h$

Volume of the new cylinder = $\pi(2r)^2 h = 4\pi r^2 h$

The ratio = $\frac{4\pi r^2 h}{\pi r^2 h} = 4$

Answer: (C) 4

Question

3. C.S.A of solid sphere is equal to

- (A) T.S.A of solid sphere
- (B) T.S.A of hemisphere
- (C) C.S.A of hemisphere
- (D) none of these

Solution

For a solid sphere, the curved surface area (CSA) is exactly the same as its total surface area (TSA), which is $4\pi r^2$.

Answer: (A) T.S.A of solid sphere

Question

4. A child reshapes a cone made up of clay of height 24 cm and radius 6 cm into a sphere, then the radius of sphere is

- (A) 24 cm
- (B) 12 cm
- (C) 6 cm
- (D) 48 cm

Solution

Volume of sphere $\left(\frac{4}{3}\pi r^3\right)$ = Volume of cone $\left(\frac{1}{3}\pi r^2 h\right)$

$$\begin{aligned}\frac{4}{3} \times \pi \times r^3 &= \frac{1}{3} \times \pi \times 6 \times 6 \times 24 \\ r^3 &= \frac{1 \times \pi \times 6 \times 6 \times 24 \times 3}{3 \times 4 \times \pi} \\ r^3 &= 6^3 \implies r = 6 \text{ cm}\end{aligned}$$

Answer: (C) 6 cm

Question

5. If the volume of sphere is $36\pi \text{ cm}^3$, then its radius is equal to

- (A) 3 cm
- (B) 2 cm
- (C) 5 cm
- (D) 10 cm

Solution

$$\begin{aligned}\frac{4}{3}\pi r^3 &= 36\pi \\ r^3 &= 36 \times \frac{3}{4} \\ r^3 &= 9 \times 3 = 27 \implies r = 3 \text{ cm}\end{aligned}$$

Answer: (A) 3 cm

Question

6. The curved surface area of a right circular cylinder of radius 1 cm and height 0.1 m is equal to

- (A) $0.2\pi \text{ cm}^2$
- (B) $20\pi \text{ cm}^2$
- (C) $2\pi \text{ cm}^2$
- (D) 2 cm^2

Solution

Radius $r = 1 \text{ cm}$, Height $h = 0.1 \text{ m} = 10 \text{ cm}$.
CSA of cylinder $= 2\pi rh = 2\pi \times 1 \times 10 = 20\pi \text{ cm}^2$.

Answer: (B) $20\pi \text{ cm}^2$

Question

7. Find the slant height if the frustum of height 15cm and having its diameters as 24cm and 8cm

- (A) 17 cm
- (B) 16 cm
- (C) 9 cm
- (D) 7 cm

Solution

Height $= 15 \text{ cm}$, Radius $r_1 = \frac{24}{2} = 12 \text{ cm}$, Radius $r_2 = \frac{8}{2} = 4 \text{ cm}$.

$$\begin{aligned}l &= \sqrt{h^2 + (r_1 - r_2)^2} \\ &= \sqrt{15^2 + (12 - 4)^2} = \sqrt{15^2 + 8^2} = \sqrt{225 + 64} = \sqrt{289} = 17 \text{ cm}\end{aligned}$$

Answer: (A) 17 cm

Question

8. If the height is 2m and the base area is 250 m^2 then the volume of cylinder =

- (A) 125 m^3
- (B) 500 m^3
- (C) $500\pi \text{ m}^3$
- (D) $125\pi \text{ m}^3$

Solution

Base Area of the Cylinder $= \pi r^2 = 250 \text{ m}^2$, Height $= 2 \text{ m}$.

Volume of the Cylinder $= \pi r^2 h = (250)(2) = 500 \text{ m}^3$.

Answer: (B) 500 m^3

Question

9. If the lateral surface area of a cylinder is 94.2 sq. cm and its height is 5 cm , then find the radius of its base. ($\pi = 3.14$)

- (A) 5 cm
- (B) 8 cm
- (C) 3 cm
- (D) 4 cm

Solution

LSA (CSA) of the cylinder $= 94.2 \text{ sq. cm}$.

$$2 \times 3.14 \times r \times 5 = 94.2$$

$$r = \frac{94.2}{2 \times 5 \times 3.14} = 3 \text{ cm}.$$

Answer: (C) 3 cm

Question

10. What is the height of a solid cylinder of diameter 10 cm and total surface area is 440 sq.cm ?

- (A) 10 cm
- (B) 30 cm
- (C) 15 cm
- (D) 9 cm

Solution

TSA of cylinder $= 2\pi r(h + r) = 440$. Radius $r = 5 \text{ cm}$.

$$2 \times \frac{22}{7} \times 5(h + 5) = 440$$

$$h + 5 = \frac{440 \times 7}{2 \times 22 \times 5} = 14$$

$$h = 14 - 5 = 9 \text{ cm}$$

Answer: (D) 9 cm

Question

11. What will be the curved surface area of a right circular cylinder having length 160 cm and radius of the base is 7 cm?

- (A) 6600 cm²
- (B) 5050 cm²
- (C) 7040 cm²
- (D) 7090 cm²

Solution

$$\begin{aligned}\text{CSA of the cylinder} &= 2\pi rh \\ &= 2 \times \frac{22}{7} \times 7 \times 160 = 7040 \text{ cm}^2.\end{aligned}$$

Answer: (C) 7040 cm²

Question

12. The ratio of the radii of two cylinders is 2:3 and the ratio of their heights is 5:3. The ratio of their volumes will be

- (A) 4:9
- (B) 25:9
- (C) 8:27
- (D) 20:27

Solution

$$\begin{aligned}\frac{\text{Volume of cylinder 1}}{\text{Volume of cylinder 2}} &= \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \left(\frac{r_1}{r_2}\right)^2 \frac{h_1}{h_2} \\ &= \left(\frac{2}{3}\right)^2 \times \frac{5}{3} = \frac{4}{9} \times \frac{5}{3} = \frac{20}{27} = 20 : 27.\end{aligned}$$

Answer: (D) 20:27

Question

13. If the ratio of the radius of two spheres is 2:5, then the ratio of their surface areas is

- (A) 4:25
- (B) 64:125
- (C) 3:5
- (D) 25:9

Solution

$$\begin{aligned}\frac{\text{Surface area 1}}{\text{Surface area 2}} &= \frac{4\pi r_1^2}{4\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 \\ &= \left(\frac{2}{5}\right)^2 = \frac{4}{25} = 4 : 25.\end{aligned}$$

Answer: (A) 4:25

Question

14. If the volume of a right circular cone is $96\pi \text{ cm}^3$ and its height is 8 cm, its slant height is

- (A) 10 cm
- (B) 13 cm
- (C) 5 cm
- (D) 25 cm

Solution

Volume of cone = $\frac{1}{3}\pi r^2 h = 96\pi$.

$$\frac{1}{3} \times \pi \times r^2 \times 8 = 96\pi$$

$$r^2 = \frac{96 \times 3}{8} = 36 = 6^2 \implies r = 6$$

$$l = \sqrt{h^2 + r^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$$

Answer: (A) 10 cm

Question

15. The number of lead balls of diameter 2 cm each, that can be made from a sphere of diameter 16 cm is

- (A) 562
- (B) 64
- (C) 512
- (D) 542

Solution

Number of lead balls = $\frac{\text{Volume of the sphere}}{\text{Volume of Lead ball}}$

$$= \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{8 \times 8 \times 8}{1 \times 1 \times 1} = 512.$$

Answer: (C) 512

Question

16. Curved surface area of solid sphere is 24 cm^2 . If the sphere is divided into two hemispheres, then the total surface area of one of the hemispheres is

- (A) 12 cm^2
- (B) 8 cm^2
- (C) 16 cm^2
- (D) 18 cm^2

Solution

TSA of hemisphere = $\frac{3}{4} \times \text{CSA of sphere}$

$$= \frac{3}{4} \times 24 = 18 \text{ cm}^2.$$

Answer: (D) 18 cm^2

Question

17. Two right circular cones have equal radii. If their slant heights are in the ratio 4:3, then their respective curved surface areas are in the ratio

- (A) 16:9
- (B) 8:6
- (C) 4:3
- (D) 3:4

Solution

$$\frac{\text{CSA of cone 1}}{\text{CSA of cone 2}} = \frac{\pi r l_1}{\pi r l_2} = \frac{l_1}{l_2} = \frac{4}{3} = 4 : 3.$$

Answer: (C) 4:3

Question

18. If the diameter and height of a right circular cone are 12 cm and 8 cm respectively, then the slant height is

- (A) 10 cm
- (B) 20 cm
- (C) 30 cm
- (D) 96 cm

Solution

$$r = 6 \text{ cm}, h = 8 \text{ cm}.$$

$$l = \sqrt{h^2 + r^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10.$$

Answer: (A) 10 cm

Question

19. If the circumference at the base of a right circular cone and the slant height are 120 cm and 10 cm respectively, then the curved surface area of the cone is equal to

- (A) $1200\pi \text{ cm}^2$
- (B) $600\pi \text{ cm}^2$
- (C) $300\pi \text{ cm}^2$
- (D) 600 cm^2

Solution

$$\text{CSA of cone} = \frac{\text{base circumference}}{2} \times \text{slant height} = \frac{120\pi}{2} \times 10 = 600\pi \text{ cm}^2.$$

(Note: Assuming the base circumference given is 120π based on the solution provided in the text).

Answer: (B) $600\pi \text{ cm}^2$

Question

20. (Numbered 2 in text) The curved surface area of a hemi-sphere is how much times the square of its radius

- (A) π
- (B) 2π
- (C) 3π
- (D) 4π

Solution

CSA of hemisphere = $2\pi r^2 = 2\pi \times (\text{radius})^2$.

Answer: (B) 2π

Chapter 8: Statistics and Probability

Question

1. Which of the following values cannot be a probability of an event?

- (A) 0
- (B) 0.5
- (C) 1.05
- (D) 1

Solution

The values can be a probability of an event for 0, 0.5, 1.

The value cannot be a probability of an event for 1.05, since $0 \leq P(E) \leq 1$.

Answer: (C) 1.05

Question

2. If the sum and mean of a data are 407 and 11 respectively, then the number of observations in the data are:

- (A) 37
- (B) 4477
- (C) 396
- (D) 418

Solution

Given $\sum x = 407, \bar{x} = 11$.

$$\bar{x} = \frac{\sum x}{n} \implies 11 = \frac{407}{n} \implies n = \frac{407}{11} = 37.$$

Answer: (A) 37

Question

3. A fair die is thrown once. The probability of getting a prime (or) composite number is

- (A) 1
- (B) 0
- (C) $\frac{5}{6}$
- (D) $\frac{1}{6}$

Solution

$S = \{1, 2, 3, 4, 5, 6\}$. Since 1 is neither a prime nor a composite number,
Required probability = $\frac{5}{6}$.

Answer: (C) $\frac{5}{6}$

Question

4. The average of first 'n' natural number is

- (A) $\frac{n(n+1)}{2}$
- (B) $\frac{n}{2}$
- (C) $\frac{n+1}{2}$
- (D) n

Solution

Sum of first n natural numbers = $\frac{n(n+1)}{2}$. Average = $\frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}$.

Answer: (C) $\frac{n+1}{2}$

Question

5. Father of Indian statistics was

- (A) Srinivasa Ramanujam
- (B) Sagundala Devi
- (C) Prasanta Chandra Mahalaobis
- (D) Kaprekar

Solution

Answer: (C) Prasanta Chandra Mahalaobis

Question

6. In a family of 3 children the probability of having atleast one boy is

- (A) $\frac{1}{2}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{4}$
- (D) $\frac{1}{8}$

Solution

B - Boy Child, G - Girl child.

Possible cases are BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG.

The probability of having atleast one boy in a family = $\frac{7}{8}$.

Answer: (D) $\frac{7}{8}$

Question

7. If the sum of 10 data values is 265 then their mean is

- (A) 27.5
- (B) 26.5
- (C) 25.5
- (D) 26.5

Solution

$$\bar{x} = \frac{\sum x}{n} = \frac{265}{10} = 26.5.$$

Answer: (B) 26.5

Question

8. An English month is selected at random in a year. The probability that it contains 31 days is.

- (A) $\frac{6}{12}$
- (B) $\frac{7}{12}$
- (C) $\frac{5}{12}$
- (D) $\frac{1}{12}$

Solution

Months having 31 days are January, March, May, July, August, October & December.

1 Year = 12 months, 7 Months have 31 days.

Required probability $P(E) = \frac{n(E)}{n(S)} = \frac{7}{12}$.

Answer: (B) $\frac{7}{12}$

Question

9. The probability that a student will score centum in mathematics is $\frac{4}{5}$. The Probability that he will not score centum is

- (A) $\frac{1}{5}$
- (B) $\frac{4}{5}$
- (C) $\frac{3}{5}$
- (D) $\frac{4}{5}$

Solution

The Probability that score centum in mathematics = $\frac{4}{5}$.

The Probability that it not score centum in mathematics = $1 - \frac{4}{5} = \frac{5-4}{5} = \frac{1}{5}$.

Answer: (A) $\frac{1}{5}$

Question

10. The probability of an impossible event is

- (A) 1
- (B) 0
- (C) $\frac{1}{2}$
- (D) -1

Solution

Answer: (B) 0

Question

11. $P(A) + P(\bar{A}) =$ _

- (A) 0
- (B) 1
- (C) -1
- (D) 0.5

Solution

Answer: (B) 1

Question

12. If $P(A \cap B) = 0.3$, $P(\bar{A} \cap B) = 0.45$ then $P(B) =$ _

- (A) 0.14
- (B) 0.30
- (C) 0.75
- (D) 1

Solution

$$P(A \cap B) = 0.3, P(\bar{A} \cap B) = 0.45$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

$$0.45 = P(B) - 0.3$$

$$0.45 + 0.3 = P(B) \implies 0.75 = P(B)$$

Answer: (C) 0.75

Question

13. If V is the variance and σ is the standard deviation then

- (A) $V = \frac{1}{\sigma^2}$
- (B) $V = \frac{1}{\sigma}$
- (C) $V = \sigma^2$
- (D) $V^2 = \sigma$

Solution

Answer: (C) $V = \sigma^2$

Question

14. For any collection of n items, $(\sum x) - \bar{x} =$

- (A) $n\bar{x}$
- (B) $(n - 2)\bar{x}$
- (C) $(n - 1)\bar{x}$
- (D) 0

Solution

$$(\sum x) - \bar{x} = n\bar{x} - \bar{x} = \bar{x}(n - 1).$$

Answer: (C) $(n - 1)\bar{x}$

Question

15. If t is the standard deviation of x, y, z , then the standard deviation of $x + 5, y + 5, z + 5$ is

- (A) $\frac{t}{3}$
- (B) $t + 5$
- (C) t
- (D) xyz

Solution

The SD of a distribution remains unchanged when each value is added or subtracted by the same quantity.
 $\sigma = t$

Answer: (C) t

Question

16. If the standard deviation of a set of data is 1.6, then the variance is

- (A) 0.4
- (B) 2.56
- (C) 1.96
- (D) 0.04

Solution

Given $\sigma = 1.6$; Variance $= \sigma^2 = 1.6^2 = 2.56$.

Answer: (B) 2.56

Question

17. If the variance of a data is 12.25, then the S.D is

- (A) 3.5
- (B) 3
- (C) 2.5
- (D) 3.25

Solution

$$\text{S.D} = \sqrt{\text{Variance}} = \sqrt{12.25} = 3.5.$$

Answer: (A) 3.5

Question

18. In a single throw of a dice the probability of getting a multiple of 3 is

- (A) $\frac{1}{2}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{6}$
- (D) $\frac{2}{3}$

Solution

$$S = \{1, 2, 3, 4, 5, 6\}, n(S) = 6.$$

$$A = \{3, 6\}, n(A) = 2.$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{6} = \frac{1}{3}.$$

Answer: (B) $\frac{1}{3}$

Question

19. If A and B are mutually exclusive events then

- (A) $P(A) \leq P(\overline{B})$
- (B) $P(A) \geq P(\overline{B})$
- (C) $P(A) < P(\overline{B})$
- (D) None of these

Solution

Answer: (A) $P(A) \leq P(\overline{B})$

Question

20. Which of the following cannot be probability of an event?

- (A) $\frac{2}{3}$
- (B) -1.5
- (C) 15%
- (D) 0.7

Solution

Since the probability value of an event always lies from 0 to 1 ($0 \leq P \leq 1$).

Answer: (B) -1.5